Chih-Hsuan (Bella) Yang

LLM Agents and Multi Agent Systems

bellayang@anl.gov | chihhsuan-yang.github.io
GitHub | Hugging Face | Google Scholar | AgentsSci | LinkedIn

Postdoctoral researcher at Argonne working on reliable large language model (LLM) agents and multi-agent systems. First author of accepted NeurIPS 2026 main-track and Findings of EMNLP 2026 papers; first author of BioTrove (NeurIPS 2024 Spotlight). Focuses on agent harnesses, evaluation, feedback, memory, and agentic AI for Science.

Research and Technical Skills

Research: Agent harnesses, multi-agent reasoning and evaluation, protocol routing, memory, preference and reward modeling, reinforcement learning from human feedback (RLHF), DPO, PPO.

Implementation: Python, PyTorch, Hugging Face, NumPy, pandas, scikit-learn, pytest, Git; reproducible benchmark and trace-analysis workflows; HPC experimentation.

Research Experience

Argonne National Laboratory

Postdoctoral Researcher | Jul 2025 - Present

- Lead research on reliable, efficient LLM-agent collaboration for scientific reasoning, connecting experimental design, protocol execution, trace analysis, and public research artifacts.
- Study when reviewer feedback changes a solver's answer and when additional collaboration is worth its inference cost. First-author studies compare matched reasoning protocols, including a 4,181-problem primary math benchmark and cross-benchmark robustness checks.
- Mentor four students and coordinate research collaborations, guiding research questions, experiments, manuscripts, and reproducible code and data releases.

Corteva Agriscience

Visiting Scientist | Sep 2024 - Jan 2025

- Trained preference models from scratch, using real experimental outcomes to ground reward signals; incorporated scientist preferences for uncertain model judgments.
- Adapted multiple language models with Direct Preference Optimization (DPO) and Proximal Policy Optimization (PPO) for domain-specific needs.

X, Google LLC

AI Resident | Sep - Dec 2023

- Developed multi-agent LLM frameworks representing stakeholders with different interests; investigated information flow, agent coordination and communication, and bidding mechanisms.

Iowa State University

AI Graduate Researcher | Jan 2020 - Jun 2025

- Co-led BioTrove, a research-grade biodiversity image and vision-language data effort. The resulting dataset contains 161.9 million images spanning approximately 366,600 species; the paper received a NeurIPS 2024 Spotlight.
- Developed molecular language modeling and multi-fidelity methods for materials discovery, alongside research in neural PDE solvers and scientific machine learning.

Selected Accepted Publications

Precise but Uncoupled: Reviewer Precision Does Not Guarantee Critique Uptake in Multi-Agent Math Reasoning
First author. NeurIPS 2026 main track, accepted.
Separates reviewer error detection from critique uptake and verified repair, showing why precise review alone does not ensure better final answers.

LLMs Can Predict Failure Risk, But Struggle to Predict Which Collaboration Protocol Pays Off: Cost-Aware Protocol Routing Across Reasoning Tasks
First author. Findings of EMNLP 2026.
Audits the gap between predicting failure and choosing a cost-effective collaboration protocol, using matched four-protocol evaluations and two solver families.

BioTrove: A Large Curated Image Dataset Enabling AI for Biodiversity
First author. NeurIPS 2024 Datasets and Benchmarks, Spotlight.
Research-grade biodiversity data, vision-language models, and zero-shot evaluation across species, life stages, and taxonomic levels.

MolGen-Transformer: A Molecule Language Model for the Generation and Latent Space Exploration of Organic Molecules
First author. Computational Materials Science 269, 114549, 2026.
Transformer-based molecular generation and controlled latent-space exploration for molecular and materials discovery.

Selected Manuscripts Under Review

Wrong but Useful: Trajectory Value Beyond Answer Correctness in Multi-Agent Messages
First author. Submitted to AAAI 2027, under review. Public preprint.

Beyond Division of Labor: How Roles Shape Feedback and Recovery in Multi-Agent Systems
First and corresponding author. Submitted to AAMAS 2027, under review.

What Is This Memory Useful For? Measuring Task-Conditioned Utility in Long-Term Agent Memory
First and corresponding author. Submitted to AAMAS 2027, under review.

Education

Ph.D., Mechanical Engineering - Iowa State University, 2025
Co-major in Computer Engineering; minor in Applied Mathematics.

M.S., Computer Science - Iowa State University, 2025

B.S., Bioenvironmental Systems Engineering - National Taiwan University, 2019

Full publications, project pages, code, and datasets: chihhsuan-yang.github.io